

Walid Bin Ataur Rahman Akash

****Design Verification (DV) Engineer at Dynamic Solution Innovators Ltd.****

Contact

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Summary

Design Verification Engineer with over 3+ years of experience in functional verification of complex digital IPs and SoCs. Strong background in SystemVerilog, UVM, SVA, and coverage-driven verification methodologies. Experienced in building reusable verification environments, developing test plans, writing constrained-random and directed tests, creating assertions, and performing debug at RTL and waveform level.

Hands-on expertise in verifying protocols and interfaces, analyzing failures, closing functional and code coverage, and ensuring design quality through regression planning and root-cause analysis. Comfortable working across the full verification lifecycle, from spec review and verification planning to testbench development, simulation debug, and signoff support. Strong collaboration with RTL design, architecture, and backend teams to improve verification efficiency and product reliability.

Technical Skills

Core Competencies

- Processor Design & Verification
- Universal Verification Methodology (UVM)
- Constrained-random verification
- Functional and code coverage closure
- Assertion-based verification
- Scoreboards, monitors, agents, sequences
- RTL debug and waveform analysis
- Regression triage and bug tracking
- Verification planning and testbench architecture
- FPGA Prototyping & Verification
- FPGA Emulation

Programming Languages & Scripting

- SystemVerilog/Verilog
- Python
- C/C++
- Cocotb
- Bash/Perl

Tools & Platforms

- ****Design/Implementation/Simulation:**** Synopsys VCS, Mentor's Questa, Xilinx Vivado, Verilator
- ****Debug/Analysis/lint:**** Synopsys Verdi, Synopsys Spyglass, GTKWave
- ****Cloud FPGAs:**** AWS EC2 F1 and F2 Instances
- ****Version Control:**** Git

FPGA Devices

- AMD Virtex™ UltraScale+™ VU19P
- AMD Virtex™ UltraScale+™ VU47P+HBM
- AMD Virtex™ UltraScale+™ VU9P

Protocols & Standards

- PCIe
- UART/SPI/I2C
- AXI4
- APB
- JTAG

Professional Experience

****Dynamic Solution Innovators Ltd.****

Total Tenure: 3 years 2 months

- ****Design & Verification Engineer****
- April 2023 - Present

Significant Project Experience

Project 1: RISC-V Vector Processor For Edge Computing

- ****Client****: California-based Fabless Semiconductor Company
- ****Description****: Developed and verified an RV64 scalar processor with a vector co-processor that is fully RVV-1.0 compliant, including custom instructions.
- ****Key Responsibilities****:
 - Applied assertions for simulation-time condition checking.
 - Developed coverage-driven verification (CDV) plans and tracked metrics.
 - Protocol verification (AXI/AHB/APB/TileLink/custom interfaces)
 - Regression management and automation
 - Designed and implemented out-of-order execution for vector instructions.
 - RTL bug root-cause analysis
 - CI-integrated verification workflows
 - Verification of IPs or subsystems in SoC environments
 - Development and maintenance of UVM testbenches
 - Creation of reusable verification components
 - Debugging corner cases, protocol violations, and reset/clocking issues
 - Coverage analysis and closure support
 - Collaboration in full chip or block-level verification flows
 - Managed FPGA RTL development, Bit stream generation, and FPGA Emulation.
 - Integrated verification environments with CI/CD frameworks.
 - Coordinated multi-team RTL releases.

Education

- ****Military Institute of Science and Technology (MIST)****
- B.Sc. in Electrical, Electronic and Communication Engineering (February 2019 - February 2023)

Publications

- ****Improved Channel Model of UAV to UAV Millimeter Wave Link Under Hovering Fluctuations and Doppler Effect****

Languages

- ****Bengali:**** Native or Bilingual
- ****English:**** Full Professional